SPY ALPHA RESEARCH

Interpretable Feature Attribution and Alpha Validation for SPY Return Dynamics

Three-way design: train → selection shortlist → one-shot final 2024 validation.
Primary IC inference uses moving-block bootstrap; rolling IC is for stability plots only.

Instrument	SPY (US equity ETF)
Sample	2016-01-01 → 2024-12-31 (hard cap; no 2025+)
Train	2016–2021 — fit + purged inner CV
Selection	2022–2023 — attribution, Borda, model compare
Final	2024 — one-shot IC / hypotheses / backtests
Scaling	Causal expanding time-series z-score
License	Proprietary Source-Available
Version	0.3.0 (research-grade hardening)

Research Report

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1. Executive summary

This report documents a two-stage research program on SPY return dynamics with an explicit **three-period** calendar. Stage A fits models on 2016–2021 and evaluates attribution on the **selection** window (2022–2023). A Borda shortlist is frozen there. Stage B evaluates information coefficients, pre-registered hypotheses, equal-weight timing, benchmarks, and grouped ablation on **final 2024 only** — never used for selection. Primary IC inference is moving-block bootstrap on the level series; Newey–West on demeaned daily contributions is secondary. Rolling IC is retained for stability plots and must not be used for significance stars.

Core principle

Forecast accuracy is secondary. The north star is **stability of explanation** and whether selected features survive as tradeable timing signals after statistical and cost-aware validation — including honest negative findings versus SPY benchmarks.

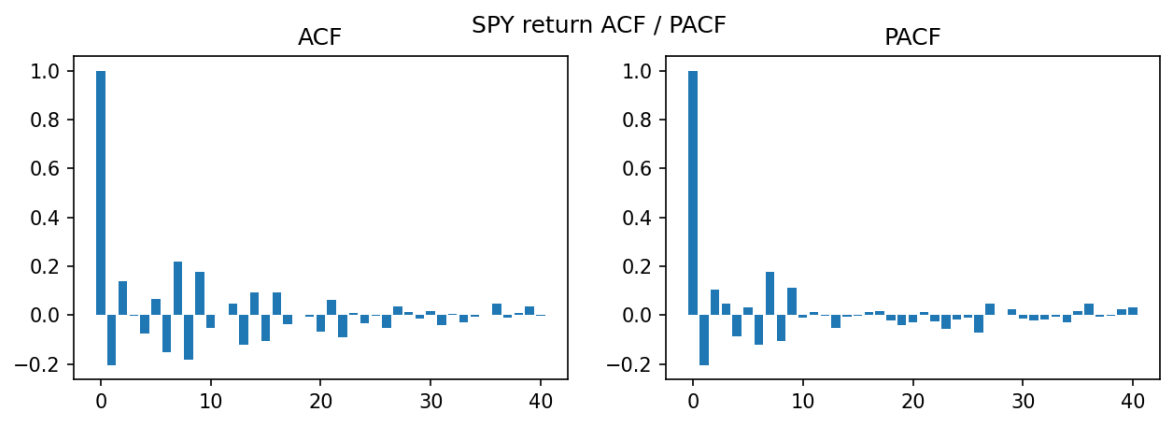
2. Data, contract, and three-way split

Features at date `t` use information known at the **close of day t**. The regression / IC target is always the forward return r_{t+1}. ExpandingScaler (and optional RollingScaler) carry training history into OOS transforms. Combined 2022–2024 analyses, if present, are labeled exploratory and are not the basis for the shortlist.

Universe	SPY OHLCV via yfinance → parquet cache
Target	1-day forward log return
Feature families	return lags, momentum, realized vol, vol-of-vol, range, volume z, calendar
Inner CV	Purged expanding walk-forward on train
Alpha shortlist	Borda consensus, top_k = 8, selection attribution only

3. Classical structure

Before machine-learning attribution, serial dependence in SPY returns is documented with ACF/PACF and an ARIMA order grid on the training window.

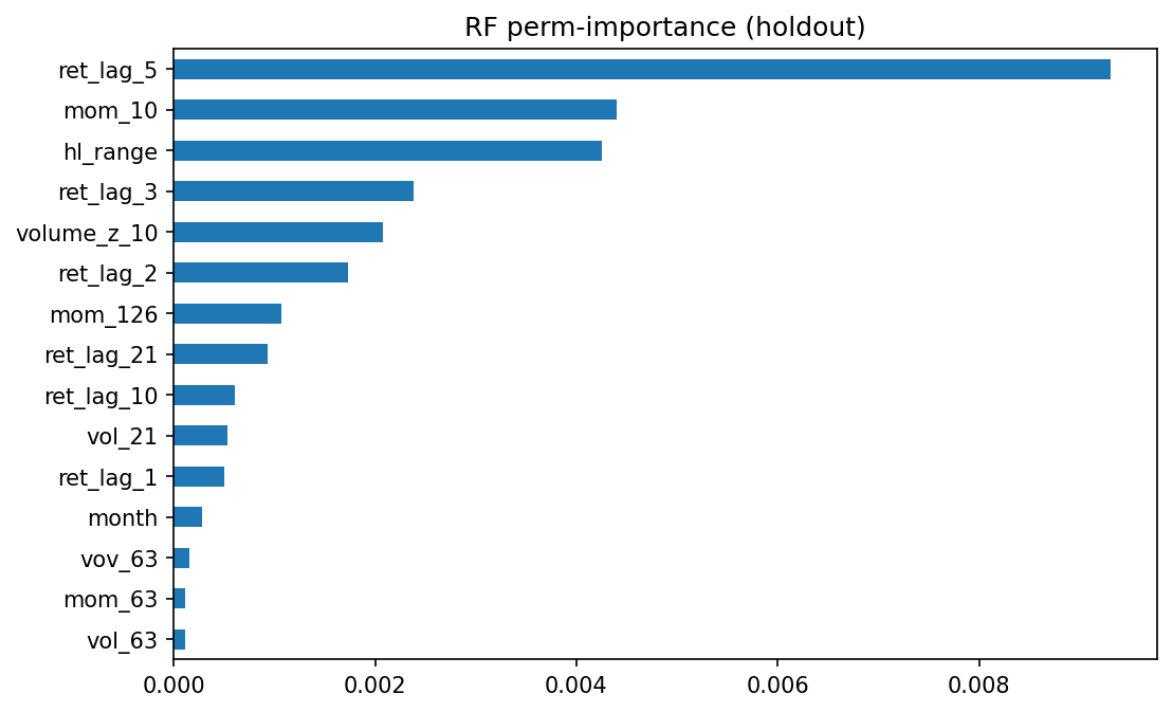


acf pacf

4. Attribution pipelines (selection window)

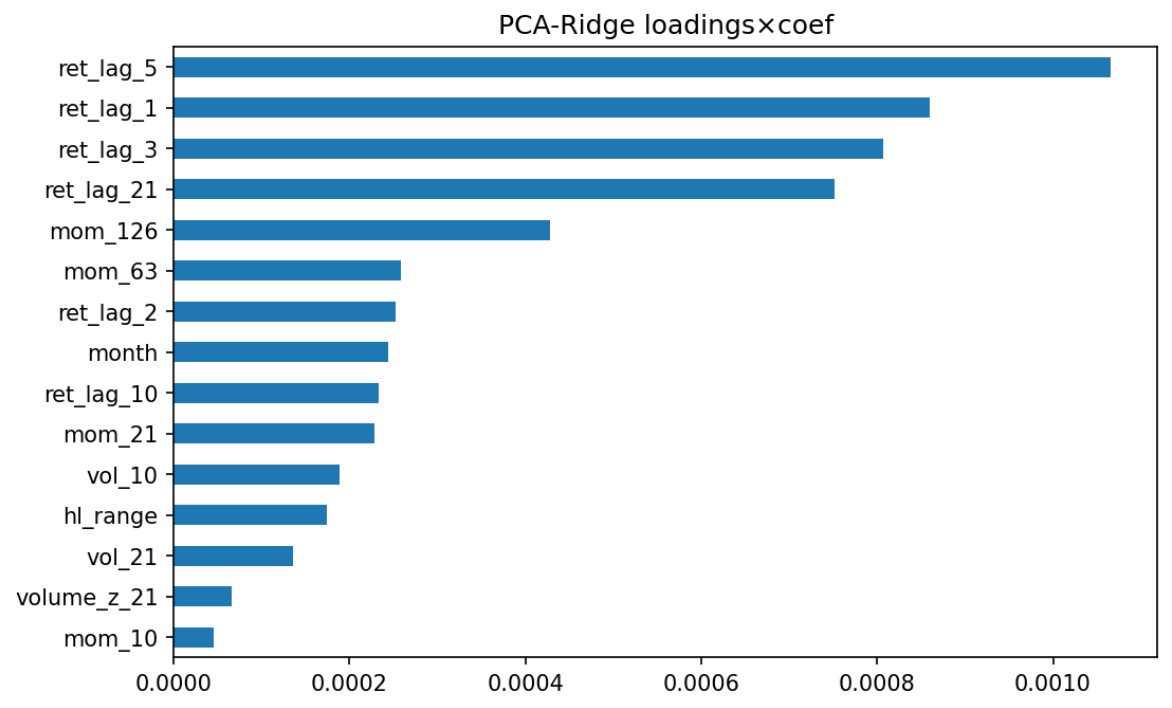
Primary attribution surfaces are computed on **2022–2023 selection** after fitting on 2016–2021. Final 2024 is untouched until Stage B.

4.1 Tree models (RF / HistGBM)

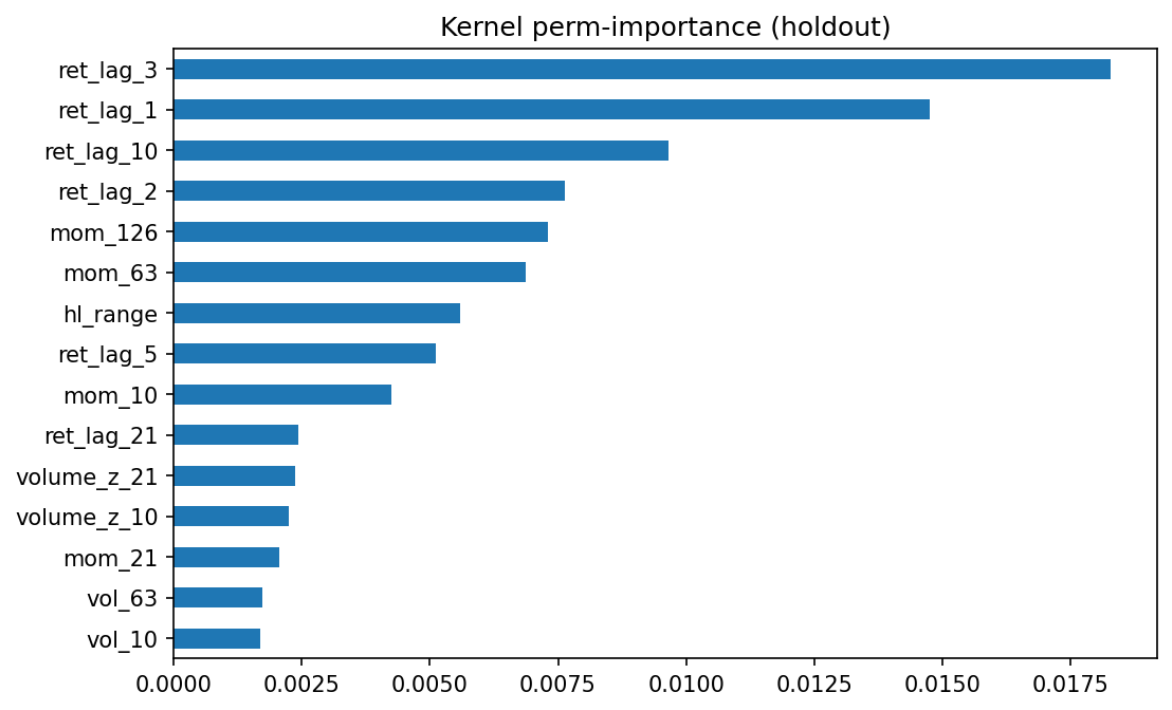


tree importance

4.2 PCA + linear / kernel

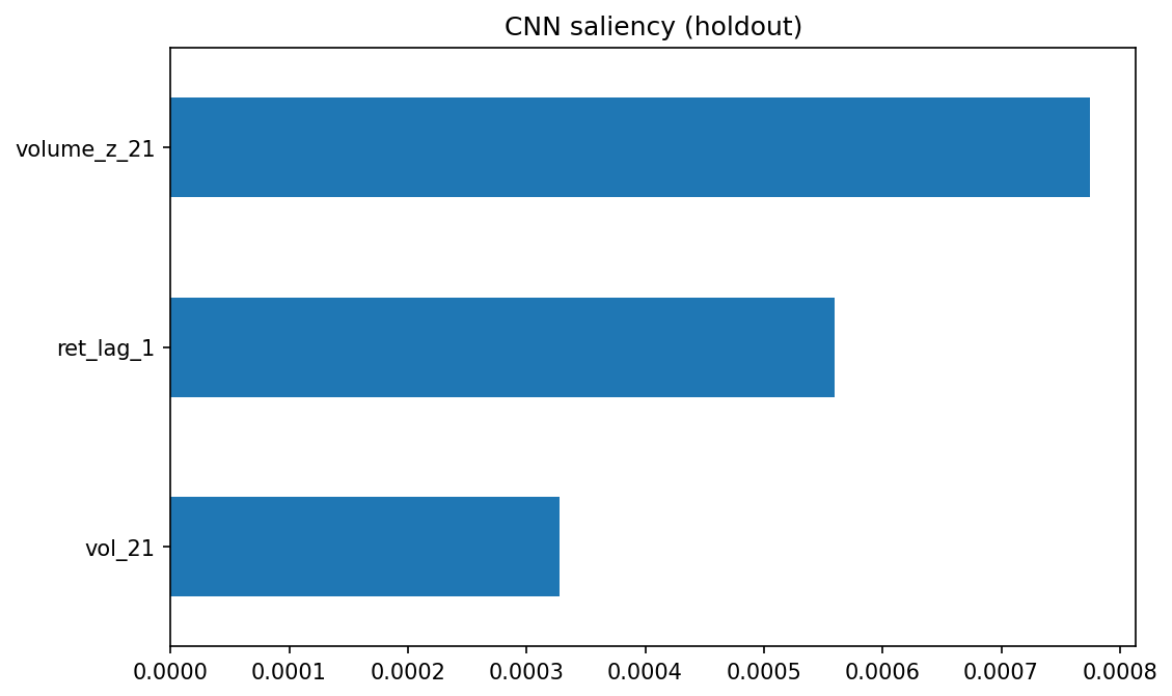


pca loadings



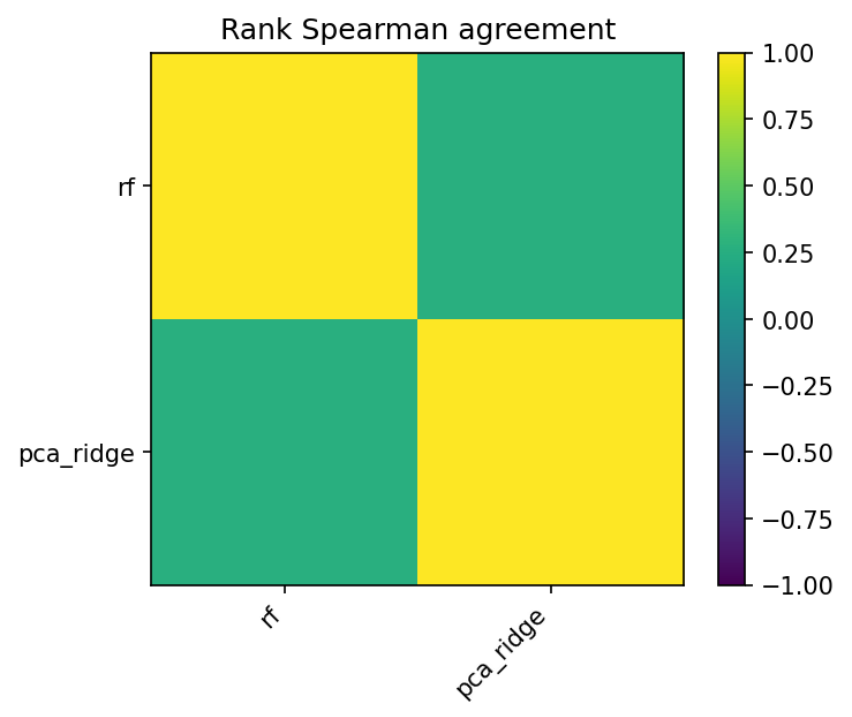
kernel perm importance

4.3 CNN feature extractor



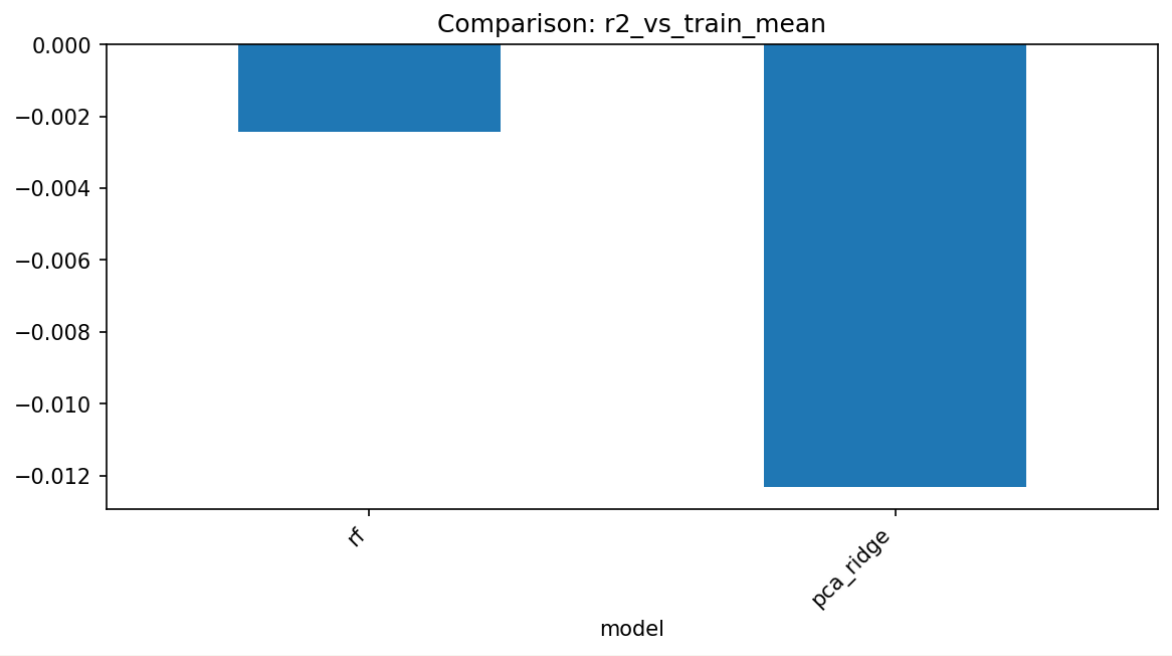
cnn saliency

4.4 Rank agreement



rank stability

4.5 Model comparison (selection)



model comparison

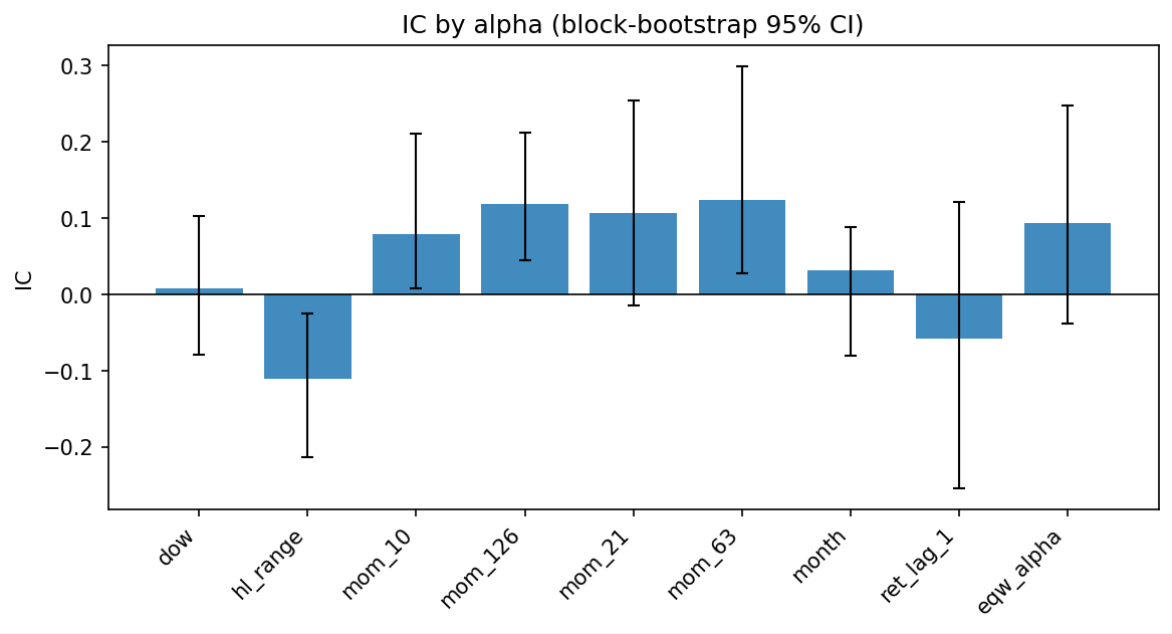
5. Consensus alpha shortlist

Selection-window importances from Random Forest and PCA-Ridge are fused with Borda consensus. The shortlist is persisted before any final-window peek and is the only input to notebooks 08–09.

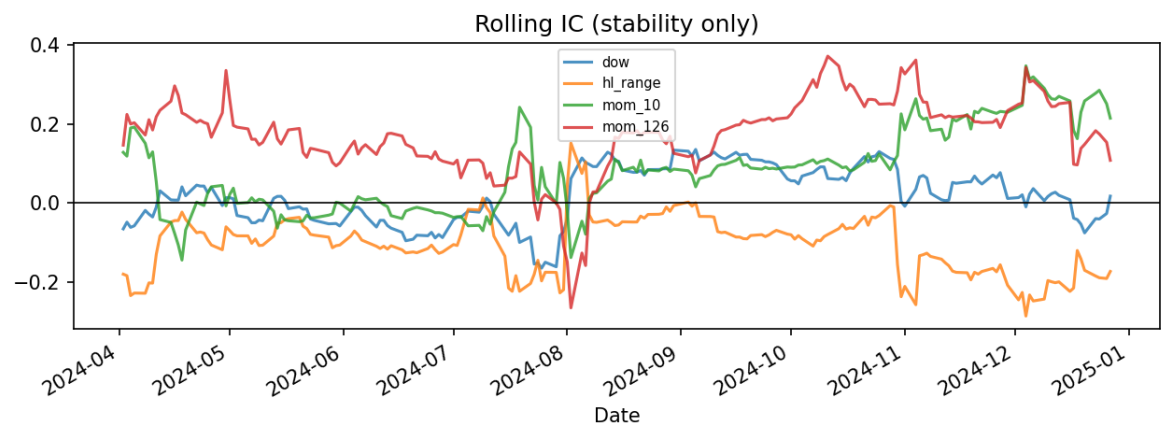
#	Selected alpha
1	dow
2	hl_range
3	mom_10
4	mom_126
5	mom_21
6	mom_63
7	month
8	ret_lag_1

6. IC, inference, and hypotheses (final 2024)

Each selected feature is mapped to a causal expanding z-score with direction fixed from the **train** IC sign. Primary inference: full-window Pearson IC with moving-block bootstrap CI and one-sided p for IC > 0. Secondary: Newey–West on the demeaned product series. BH-FDR is applied to bootstrap p-values. Pre-registered tests: H1 mean IC(alpha) > 0; H2 mean IC(equal-weight) > 0; H3 IC(equal-weight) ≥ IC(best single) via paired block bootstrap.



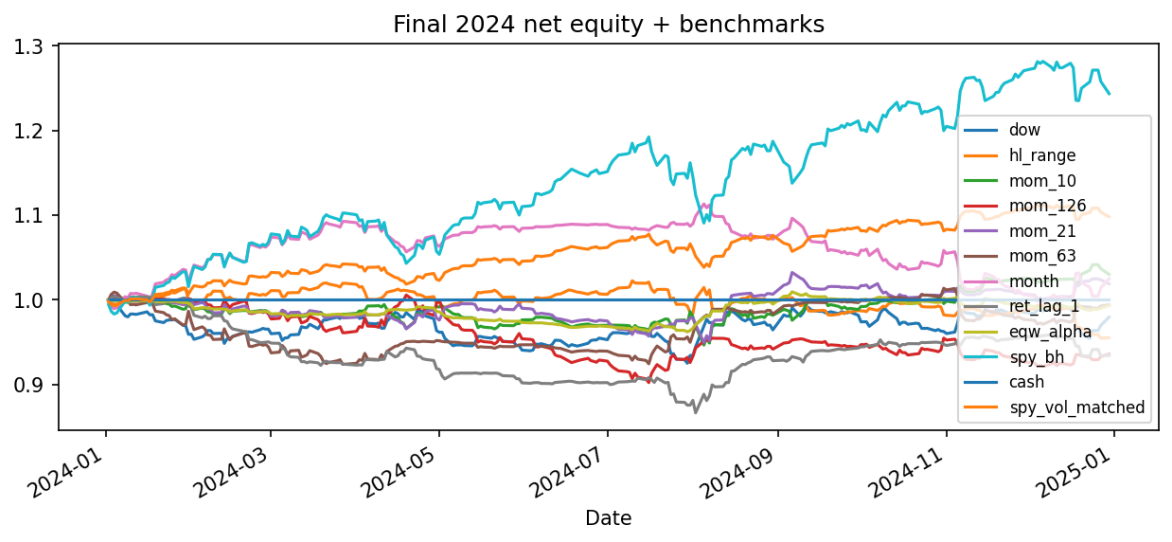
ic by alpha



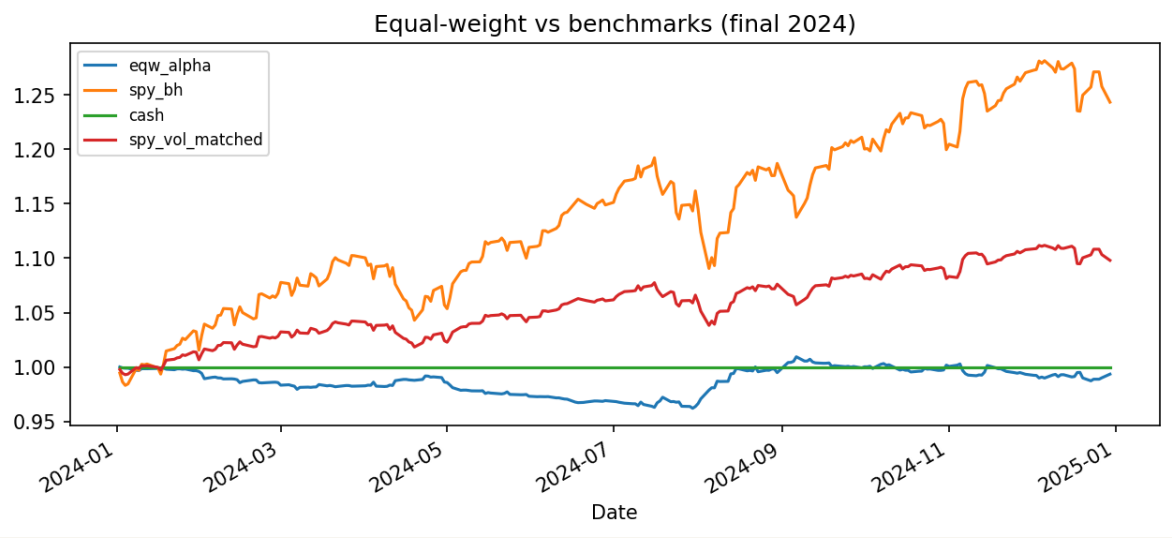
rolling ic stability

7. Final backtest, benchmarks, ablation

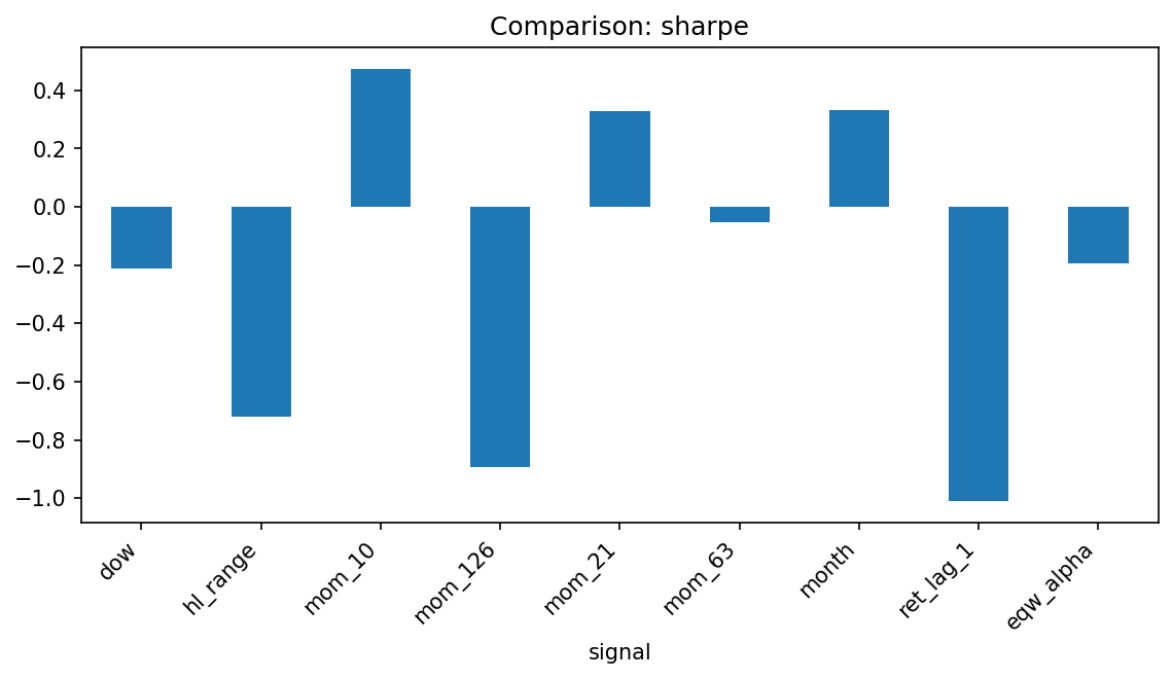
Equal-weight and single-alpha timing use identical `signal_lag` and costs. Benchmarks on final 2024: SPY buy-and-hold, cash, and volatility-matched SPY with scale fit on **selection only**. Grouped ablation drops momentum / lag / vol / volume / calendar blocks from the shortlist.



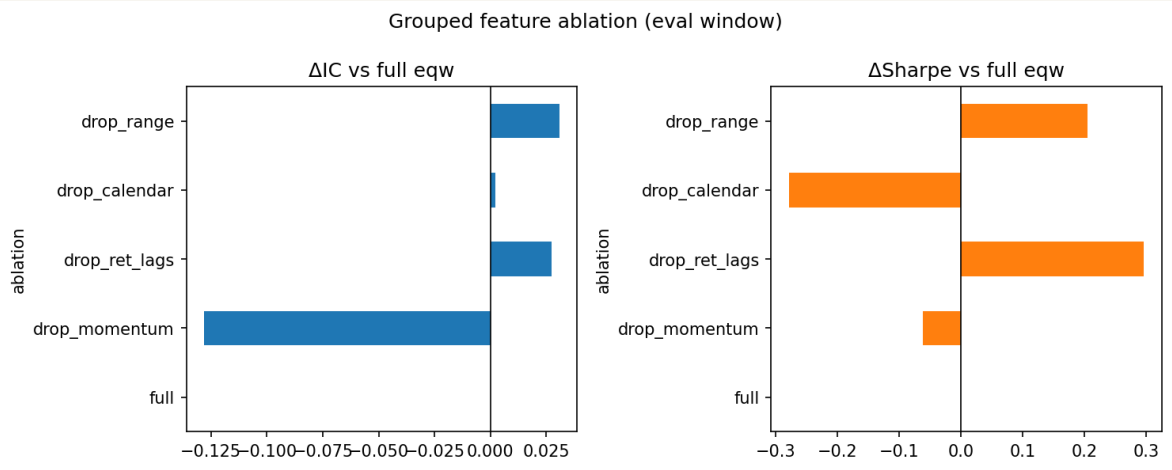
eqw vs singles equity



benchmarks equity



all signals compare



ablation groups

8. Findings status and caveats

Shortlist membership was chosen on **selection**, not cherry-picked on final Sharpe. Negative findings are first-class: equal-weight need not beat singles or SPY benchmarks. Evidence labels below are the interpretive contract for this sample — recompute after re-running notebooks 08–09 before citing numbers externally.

Claim	Effect	Evidence
Shortlist mean IC > 0 (final 2024)	Point IC + block-bootstrap 95% CI	mixed — some moms supported; calendar/lags weak
Equal-weight IC > 0 (H2)	Bootstrap one-sided p on eqw	needs more data — not decisive on 2024 alone
Equal-weight beats best single (H3)	Paired bootstrap ΔIC	not supported — singles can dominate eqw

Eqw timing vs SPY B&H / vol-matched	Final Sharpe / equity overlay	not supported — eqw may trail SPY benchmarks
Feature-group necessity (ablation)	ΔIC / ΔSharpe vs full eqw	exploratory — see ablation_groups figure

Caveats

- Prior joint 2022–2024 analyses (if any residual) are exploratory only — selection bias history.
- Final 2024 is a single calendar year; power is limited — many claims correctly stay 'needs more data'.
- SPY-only univariate timing; no cross-sectional L/S book or multi-asset confirmation.
- Costs, lag, and position map are assumptions; results are sensitive to those knobs.
- This hardening does not claim elite-desk production readiness.

Reproducibility checklist

- Three-way dates in configs/experiment.yaml (final never used for shortlist)
- Causal scaler + feature timestamp assert in notebook 02
- Selection attribution in notebooks 04–07; final-only 08–09
- Block-bootstrap IC primary; rolling IC plots only
- Benchmarks: SPY B&H;, cash, vol-matched (scale on selection)
- pytest + requirements.lock + restart Jupyter kernel after pulls
- Proprietary Source-Available LICENSE in repository root

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